

Dynamic Signature Verification through the Longest Common Subsequence Problem and Genetic Algorithms

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Abstract—The following paper includes an automated system that shows the use of signature data for biometric verification. The verification uses the dynamic distinctiveness of the signature, thus this method lies within those referred as dynamic or on-line signature verification. This proposal makes use of a variation of the Longest Common Subsequence Problem together with Genetic Algorithm for this purpose.

I. INTRODUCTION

NOWADAYS, the most frequent method to verify a person identity is the comparison of signatures. This method has several advantages, being some of these: it is simple of performing, the signature is a unique feature that is mechanized as time goes by, and its user acceptance is very extensive. However, in practice it is common that the person analyzing a signature has a minimal, if not nil knowledge on how to discern if a signature is from a person or not. On the other hand, the increasing number of transactions that imply exchange of information among individuals, makes it desirable to facilitate and improve the mechanisms that allow us to certify the identity of an individual involved in that operation.

Consequently, it would be desirable in any process where the signature is used for verification to have an expert's opinion. For the verification through a signature, two practices exist. The first one is called *offline verification* or *static testing*. This is the common form of verification and consists in comparing two firms (images) already stamped, one of which is known to be authentic and has been previously captured in the presence of someone who attests the authenticity of it. This type of method presents two major problems: the signature changes over time¹ and a signature can be *forged* if an image of an original signature is available.

Alternatively, it is possible through a number of different devices such as PDAs, some computers and different digitizers to obtain information regarding dynamic

characteristics associated to a signature. This information is much richer, since from it you can rebuild the image associated with a signature, but also other data are known such as the number of strokes and their time and direction, the pressure, angle, among others. This information helps to reduce the aforementioned problems of the offline verification, since it is possible to continuously update the pattern associated to a signature. But more importantly, it makes it harder to forge a signature since even if a copycat counts with a copy of an authentic signature, and has had opportunity to practice it, is probable that the dynamics of the fake and the authentic signature do not match. This process is known as *dynamic* or *on line* verification, being this proposal a work in this area. Some related works are those of Nalwa [1], Shimke, *et al.* [2] and recently those of Fierrez, Ortega-Garcia, *et al.* [3],[4],[5]. On the other hand, some Computer Intelligence Techniques, specifically Artificial Neural Networks [6], have been explored to solve this problem. This proposal explores the use of Genetic Algorithms for this purpose.

For a dynamic verification, two major steps are considered: *learning* and *verification*. Here it will be shown how to solve these as a variation of the *Longest Common Subsequence Problem* [7]. To do so, the paper is organized as follows. In section II the Longest Common Subsequence Problem is introduced. Section III includes how to adapt it to the Signature Learning and Verification Problem. Section IV refers how to use the Genetic Algorithm to this problem. Section V includes the results, and Section VI refers to the Conclusions and Further Work.

II. THE LONGEST COMMON SUBSEQUENCE PROBLEM.

The Longest Common Subsequence Problem (LCSP) is defined as follows. Let Σ be a finite set called an *alphabet* and $X \in \Sigma^m, Y \in \Sigma^n$ be two sequences (words) over Σ with $m \leq n$. The *length* of a word refers to the number of elements comprising it. Now Z is called a *subsequence* of X , if there is a sequence $(i_1, \dots, i_k)$, with $1 \leq i_1 < \dots < i_k \leq m$, such that $Z_j = X_{i_j}, j = 1, \dots, k$. Now, Z is a *longest common subsequence (LCS)* of X, Y if for every Z' a (common) subsequence of X, Y , the length of Z is greater or equal to the length of Z' . Thus, the LCSP obtains (the length of) a LCS for a couple of words. While

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¹ The variations of a signature through time are a problem because the stored signature used for verification is static, making in some cases very difficult to identify a person.

LCSP is an NP-hard problem [8], when the number of words is 2, the problem can be solved in polynomial time through Dynamic Programming [9]. A thorough full explanation of the LCSP and Dynamic Programming can be found in Cormen, *et al.* Book [7]. To calculate the length of a LCS for 2 words, a recursive formula can be used to obtain a matrix $L \in \mathbb{N}^{m+1 \times n+1}$, being the value of its elements defined as:

$$l_{i,j} = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ l_{i-1,j-1} + 1 & \text{if } i, j > 0 \text{ and } x_i = y_j \\ \max\{l_{i,j-1}, l_{i-1,j}\} & \text{if } i, j > 0 \text{ and } x_i \neq y_j \end{cases} \quad (1)$$

Now, if this recursion is solved in a bottom-up manner this gives an algorithm with complexity $\Theta(mn)$.

III. SIGNATURE LEARNING AND VERIFICATION AS A VARIATION OF THE LCSP.

Before explaining the learning and verification process, it is necessary to refer how the data from a signature is stored. First of all, the capturing process is done by a PDA, a digitizer, etc. These devices generate a number of *events*. An event occurs every time the data related to the pointing device² changes. These data refer to the time when an event occurred, the pressure at that moment, the current coordinates, etc. Therefore, a signature S can be thought as a succession of m events related to the signature process. Consequently, it is possible to describe $S \in \mathbb{N}^{\delta \times m}$ with δ the data cardinality related to each event.

A. The Learning Process

In an on-line signature verification process, the learning process could be recast as finding a solution $Z^* = LCS(S_1, \dots, S_k)$. Then Z^* can be used to measure if a new signature S' relates to $S_1, \dots, S_k$ by calculating the $LCS(Z^*, S')$.

Thus, let $S_1, \dots, S_k$ be k words over Σ . It was aforementioned that the LCSP is polynomial for $k = 2$. However, this is not true for every k . A simple approximation to the solution of the $LCS(S_1, \dots, S_k)$ solves:

$$\begin{aligned} Z &= LCS^*(S_1, S_2, \dots, S_k) \\ &= LCS(S_1, LCS(S_2, \dots, (LCS(S_{k-1}, S_k)) \dots)) \end{aligned} \quad (2)$$

Notice that Z is a common subsequence of $S_1, \dots, S_k$; however, it is not necessarily the longest one. On the other

hand, for the *LCSP* it is assumed that Σ is a finite set. However, for the signatures we have that, $\Sigma = \mathbb{N}^l$. Thus, a modification of (1) is necessary to compare two signatures S, S' . In this case, the following proposal was used:

$$l_{i,j} = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ \min \begin{cases} l_{i,j-1} + \|S_{:,i}\|_2, \\ l_{i-1,j-1} + \|S_{:,i} - S'_{:,j}\|_2, \\ l_{i-1,j} + \|S'_{:,j}\|_2 \end{cases} & \text{otherwise} \end{cases}, \quad (3)$$

where $S_{:,i}, S'_{:,j}$ refer to the i th column of S and the j th column of S' respectively. Clearly, this modification admits the usage of any other measure function instead of the Euclidean norm $\|\cdot\|_2$. Notice that by using (3), the difference between S, S' is obtained. In particular, if $S = S'$, then $l_{m,m} = 0$. Moreover, to obtain Z from (2) and (3), it must be noticed that in the general problem, $Z_{:,k}$ is obtained by backtracking the matrix L from $l_{m,n}$ and adding those elements for which $S_{:,i} = S'_{:,j}$. However, for this case, the probability that $S_{:,i} = S'_{:,j}$ tends to zero. So, for the purpose of this research, Z will be calculated by backtracking the matrix L from $l_{m,n}$ and considering those elements for which:

$$l_{i-1,j-1} + \|S_{:,i} - S'_{:,j}\|_2 = \min \begin{cases} l_{i,j-1} + \|S_{:,i}\|_2, \\ l_{i-1,j-1} + \|S_{:,i} - S'_{:,j}\|_2, \\ l_{i-1,j} + \|S'_{:,j}\|_2 \end{cases}. \quad (4)$$

Therefore, $Z_{:,k}$ can be calculated as a combination³ of $S_{:,i}, S'_{:,j}$. Consequently Z contains a set of characteristics common to the data hold in the valid signatures $S_1, \dots, S_k$. This data can be used to verify if a new signature is valid as follows.

B. The Verification Process.

Let $Z = LCS^*(S_1, S_2, \dots, S_k)$ be a learned signature. This sequence can be used to measure if a new signature S' is valid by solving $LCS(S', Z)$. To do so, let $d^* = l_{m,n}$. The signature S' is considered as *valid* if $d^* \leq \theta$, where θ is a threshold, whilst S' is considered as *invalid* otherwise.

By fixing θ two problems arise: a False Rejection (FR) and a False Acceptance (FA). A FR refers to the fact that $S', S_1, \dots, S_k$ are all signatures of the same individual; still

² Like a stylus or a mouse or some other element.

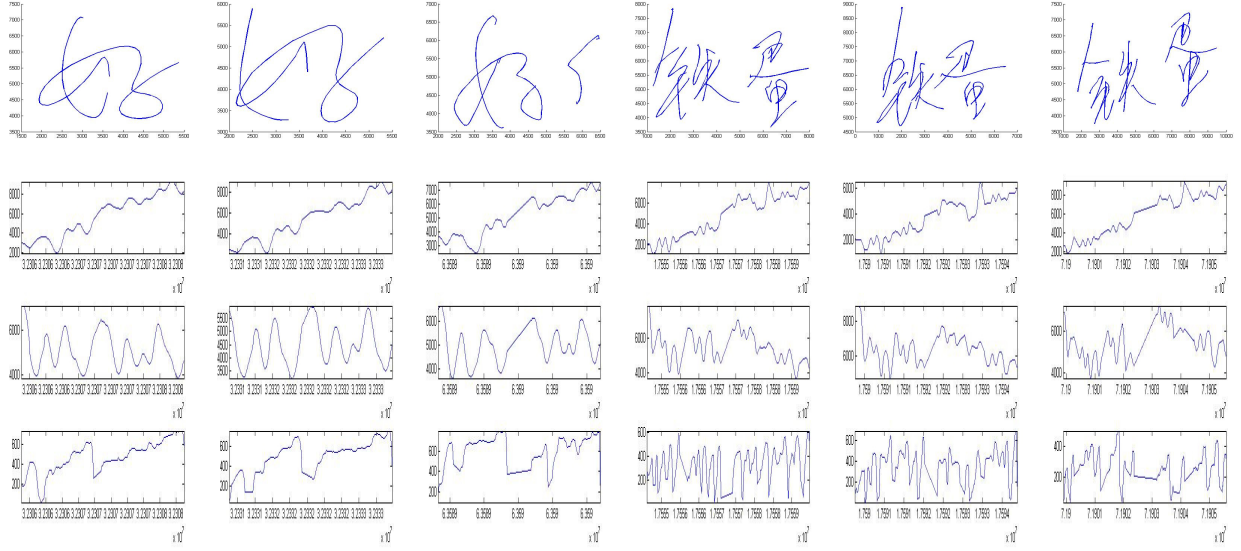


Fig. 1. Columns 1, 2, 4, 5 contain two original signatures (of two persons), whilst columns 3, 6 are forgeries. The first row contains the signature picture, and rows 2, 3 and 4 correspond to x , y , p respectively.

$d^* > \theta$.

On the other hand, let S' be a signature from a different user than the one that made signatures $S_1, \dots, S_k$, whilst $d^* \leq \theta$. This is called a FA, and S' is considered a *forgery*. There are two kinds of forgeries: random and skilled forgeries. A random forgery refers to a casual impostor producing a signature S' . The casual impostor has no information related to $S_1, \dots, S_k$. On the other hand, a skilled forgery uses information on this set to attain S' , increasing in this way the probability to accept S' as valid.

From the previous information it is clear that the selection of the threshold θ is critical. If θ is set too low, then the FA Rate (FAR) is also low; however, the FR Rate (FRR) will be high. On the other hand, by setting θ too high, the FRR will be low, but the FAR will increase. To analyze the effectiveness of an identification algorithm, a common test is to use a set that can be partitioned onto valid signatures and forgeries. From these data, the density functions of the FR/FA related to the threshold θ can be obtained. These functions correspond to a normal density and for some value of θ , these quantities are equal. This is

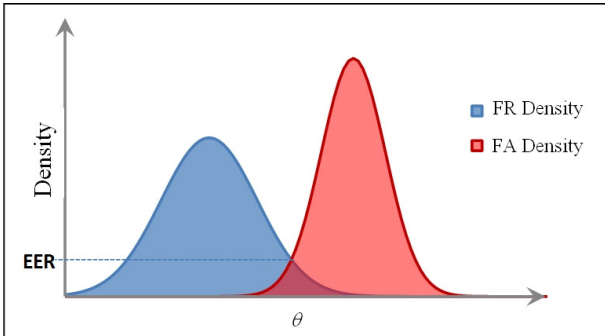


Fig. 2. Density functions of FR/FA. From these, the EER is determined.

called the *Equal Error Rate* (ERR) and is depicted in Figure 2.

One problem still to be solved refers to the fact of which combination of the data directly/indirectly retrieved by the pointing device will be used for a signature, and consequently for learning and verification processes. In general, the data directly obtained refers to x -coordinate, y -coordinate, time stamp, button status, azimuth angle, altitude angle and pressure. However, from these, it is common to calculate their derivatives and some other composite info like the distance. Experimentally, most of the proposals for online verification end up working with the (x, y) -coordinates and the pressure. Still, it is valid to ask if there is a combination including some other data that could enhance the system output.

For this purpose, we propose a *weighted version* of (3), as follows:

$$l_{i,j} = \begin{cases} 0 & \text{if } i \text{ or } j = 0 \\ \min \left\{ \begin{aligned} & l_{i,j-1} + \sqrt{\sum_{d=1}^{\delta} (\alpha_d S_{d,i})^2}, \\ & l_{i-1,j-1} + \sqrt{\sum_{d=1}^{\delta} (\alpha_d (S_{d,i} - S'_{d,j}))^2}, \\ & l_{i-1,j} + \sqrt{\sum_{d=1}^{\delta} (\alpha_d S'_{d,j})^2} \end{aligned} \right\} & \text{otherwise} \end{cases} \quad (5)$$

where $\alpha_i \geq 0, \sum_i \alpha_i = 1$. Now, the problem of selecting the best collection of data is equivalent to selecting the best vector α . To do so, we propose the following Genetic Algorithm approach.

IV. GENETIC ALGORITHM

Genetic Algorithm (GA) is a knowledge model inspired in the evolution mechanisms observed in nature. GA usually uses the following cycle [10],[11]:

- Generation of an initial population in a random way.
- Evaluation of the fitness through some objective function of every individual that belongs to the population.
- Generation of a new population through operations like crossover and mutation applied to best fitted individuals.
- Using the newly obtained population iterate until the termination criterion is fulfilled or a certain number of generations is reached.

The initial population is represented by a matrix $A_1 \in \mathbb{R}^{n_s \times \delta}$ where n_s is the number of individuals, and the sum of every row of A_1 is equal to one. Thus, a row from matrix A_1 represents an individual, and this individual corresponds to a vector α used in (5). The detailed description of the rest of these processes follows.

A. Selection Process

To explain the selection process let $g=1, \dots, n_g$, where n_g is the total number of generations, let us introduce the matrix F_g which is the set of parents of a given population while A_g represents the current population. F_g is a submatrix of A_g where $A_g \in \mathbb{R}^{n_s \times \delta}$, $F_g \in \mathbb{R}^{n_p \times \delta}$, and n_p is the number of parents of the population.

The Selection Process S_g computes the fitness of each individual and selects the best n_p individuals from A_g . For this case, the fitness of each individual corresponds to the EER and Tournament as the Selection Method.

B. Crossover

Let $C(F_g)$ be the crossover operator which can be defined as the combination of the parent's information. For this research, we did a convex combination from a couple of parents $(F_{i,:})_g, (F_{j,:})_g$ by using a random number $\gamma \in [0,1]$.

Thus, if $(S_{i,:})_g$ is the son from $(F_{i,:})_g, (F_{j,:})_g$, then:

$$(S_{i,:})_g = \gamma(F_{i,:})_g + (1-\gamma)(F_{j,:})_g. \quad (1)$$

C. Mutation

The mutation operator changes some elements of an individual in a random way from a fixed probability factor P_m ; in other words, we vary the components of some *genes*. This operator is extremely important, because it assures the maintenance of the diversity inside the population, which is

basic for the evolution. The problem here is that we need to guarantee that the sum of the genes of an individual α remains one. Thus we proposed the following approach:

$$\begin{aligned} \alpha_i &= \alpha_i + \gamma, \\ \alpha_j &= \alpha_j + \gamma/(\delta-1), 1 \leq j \neq i \leq \delta \end{aligned} \quad (2)$$

where $\gamma = \text{rand}(0, \alpha_{\min})$ being $\text{rand}(a,b)$ a function that generates a random value within the range $[a,b]$ with uniform distribution and $\alpha_{\min} = \min\{\alpha_i : 1 \leq i \leq \delta\}$.

The new generation is obtained from the mutation of the parents selected, their offspring, and the best solution from A_g . The GA process ended when the total number of generations n_g is reached.

V. RESULTS

To test the system, a Database with signatures from a number of individuals is needed. Even more, for each individual a group of counterfeiters is required. This is necessary to assess the "vulnerability" of the system against attempts to forge a signature from the FRR. From these data, the FAR and the FRR is obtained.

Unfortunately, there are few databases with the necessary information to verify the authenticity of a signature. One possibility is the MCYT-100 [12], which is probably the best DB for this purpose. Unfortunately, the European Union imposed new conditions for sharing of biometric data, which is the case of signatures, thus the access to MCYT-100 has been temporarily disabled.

Another available DB is the one used during the Signature Verification Competition 2004 (SVC04) [13]. A reduced set from it is available for testing. In it, there are signatures of 40 users and 40 samples of every person. The first 20 signatures are original and were taken in two sessions (10 in each session). The remaining 20 are forgeries. It is worth mentioning that the signatures were not genuine signatures of the user, and for the forgeries, the forger was informed on how good was her/his "fake" by letting the forger know about the signature dynamics and even allowing she/he to practice.

On the other hand, every event associated to a signature has 13 data: current position x, y , time t , button status b , azimuth a , altitude h , pressure p , the distance and the derivatives of x, y, a, h and p . These data were normalized to the range $[0,1]$.

To calculate the EER, a number of test were done using the SVC04 DB. First, from the first 10 original signatures⁴, $k=2,3,5$ sample signatures $S_1, \dots, S_k$ were randomly

⁴ These signatures were those captured during the first session.

Table I
Mean values from the best solutions obtained by GA.

Mean Values $(x, y, t, bs, a, h, p, dist, \dot{x}, \dot{y}, \dot{a}, \dot{h}, \dot{p})$	
SL, $k = 2$	(0.191, 0.178, 0.092, 0.010, 0.010, 0.016, 0.052, 0.192, 0.107, 0.103, 0.007, 0.005, 0.037)
CL, $k = 2$	(0.186, 0.191, 0.090, 0.009, 0.011, 0.014, 0.051, 0.191, 0.116, 0.091, 0.007, 0.006, 0.036)
SL, $k = 3$	(0.178, 0.180, 0.085, 0.010, 0.010, 0.014, 0.051, 0.180, 0.099, 0.136, 0.005, 0.007, 0.044)
CL, $k = 3$	(0.185, 0.183, 0.086, 0.010, 0.010, 0.014, 0.053, 0.173, 0.106, 0.126, 0.009, 0.008, 0.038)
SL, $k = 5$	(0.189, 0.183, 0.088, 0.010, 0.010, 0.014, 0.052, 0.178, 0.113, 0.109, 0.007, 0.006, 0.041)
CL, $k = 5$	(0.169, 0.171, 0.082, 0.010, 0.010, 0.014, 0.047, 0.167, 0.130, 0.151, 0.009, 0.007, 0.033)

picked for the learning process.

For the learning process, two schemes were proved. The first one obtained Z directly from (5). A second process, first calculated Z from (5), and then Z was used within an iterative process until the learned pattern converges. This was done in the following manner.

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conv_learning ( $S_1, \dots, S_k$ )
1.  $Z = LCS^* (S_1, \dots, S_k)$ 
2. Let  $i = 1$ 
3. repeat
4.      $Z' = Z$ 
5.      $Z = LCSP (S_i, Z)$ 
6.      $i = (i + 1) \pmod k$ 
7. until ( $Z' != Z$ )

```

To test the system, every user signature (original and fakes) besides those employed in the learning process was used. This process was tested 10 times for every scenario over 100 generations with populations of 64, 96 and 128 individuals, the data and the samples was normalized; having 128 as the best sampling rate and also as the best population. The mean values from the best solutions resulting from the GA approach are summarized in Table I, whilst the mean of the results for the EER are reported in Table II.

Table II
BEST EER OBTAINED FOR THE DIFFERENT SCENARIOS TESTED.

	$k = 2$	$k = 3$	$k = 5$
Straight learning EER (SL)	17.32%	15.18%	12.60%
Learning EER w/convergence (CL)	13.85%	12.27%	10.63%

Regretfully, these results are still far from those obtain by the winner of the SVC04, which obtained an EER of 6.9%. However, they are approximately the same of those who used a LCSP approach. One way to improve these results might be to include some other techniques like the ones mentioned in [14].

Conclusions and Future Work

The results obtained in this investigation show, as has already been established by other authors, the feasibility for the use of signatures for the identification of persons.

The results included here can be improved taking into account the proposals of other researchers in the field relating to the normalization of the similarity function for different users [14], as well as the combination of different similarity functions to build a more robust [6].

From the data shown in Table I, we can establish that the number of samples used for the learning process was approximately the same for every k , and that the best data is associated, to the (x, y) -coordinates, or some other data obtained from it (like the distance or their derivatives). Also the pressure plays an important role. This is consistent with the proposals of the works in this area; since more than

90% is associated to this data. However, most of the researchers omit the time data as it is intrinsically contained as an “index” of the signature events. Nevertheless, these experiments consistently assigned a value close to 9% to the time. So these experiments point out that time should not be omitted so easily from the calculations.

In the near future, we intend to create a DB using the proposed platform, which also has the participation of people not only for their own signatures, but to allow them to participate in a guided process of counterfeiting. In the GA topic, we would like to test others Selection Methods such as Roulette, and Elitism, and different operators and variations. We believe this would be of practical use as we intend to integrate some other models and distance functions used for the signature verification such as the Hidden Markov Chains [3, 4] and the Levenhstein function [2].

An interesting test that was not done here, but that could be included later, would be to ask the opinion of two sets of people who have to carry out the task of identifying signatures. One of these groups should be formed by people who have an informal instruction and another group of people with a formal instruction in the identification of signatures (graphologists). Their opinions would be used to evaluate the performance of the system against their human counterparts.

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